

ECG Phase Estimator for Cardiac Artifact Removal

Extracted from: Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Problem Formulation (Background)

Following repetitive-control modeling conventions, the measured ear-EEG signal is modeled as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *The beat period T_k is assumed slowly varying, dominated by respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.*

The repetitive disturbance is represented as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Problem 1. *Estimate $\hat{t}_k$ robustly so that EEG samples can be phase-aligned via the phase model. The estimated cardiac phase between beats advances as*

$$\hat{\phi}(t) = \frac{t - \hat{t}_k}{\hat{T}_k}, \quad \hat{T}_k = \hat{t}_{k+1} - \hat{t}_k, \quad (4)$$

where $\hat{T}_k$ is the estimated beat period.

Note 1. *Since T_k is slowly varying (Assumption 1), $\hat{T}_k$ can be reliably predicted between successive R-peaks. The PI tracker advances phase continuously between beats so that $\hat{\phi}(t)$ remains well-defined in real-time causal operation.*

1 ECG Phase Estimator

1.1 Min/max tracker + PI/PID phase tracking

A QRS-emphasis band-pass feeds two streaming trackers: a peak tracker $pk(t)$ (fast attack, slow release) and a baseline tracker $bl(t)$ (slow moving average). The normalized gate

$$g(t) = \frac{x(t) - bl(t)}{pk(t) - bl(t)} \quad (5)$$

is amplitude-drift invariant, so posture/contact drift does not defeat detection. A Schmitt hysteresis and a 250 ms refractory yield clean beat gating; the fiducial t_k is the raw-reference maximum within the gated window (low jitter).

A PI phase-locked loop converts sparse fiducials into a continuous phase. Between beats the phase advances by the instantaneous rate estimate,

$$p(t+1) = [p(t) + \text{fr}(t)] \bmod 1, \quad \text{fr}(t) = \frac{1}{\text{RR}_{\text{est}}(t) f_s}, \quad (6)$$

where f_s is the sample rate. At each fiducial, a proportional term realigns phase and an integral term updates the rate estimate,

$$p \leftarrow p - K_p e_p, \quad (7)$$

$$\text{RR}_{\text{est}} \leftarrow \text{RR}_{\text{est}} + K_i (\text{RR}_{\text{meas}} - \text{RR}_{\text{est}}), \quad (8)$$

with e_p the wrapped phase error and K_p, K_i the PI gains. The loop bandwidth is set above respiratory sinus arrhythmia (RSA, ~ 0.2 – 0.4 Hz) and below per-beat detection noise. An arrhythmia supervisor classifies each beat by the ratio $\text{RR}_{\text{meas}}/\text{RR}_{\text{est}}$ and applies the matching reflex—*premature*: trust the measured phase, freeze the rate and learning; *missed*: trust the prediction, keep the rate—protecting both the loop and the template.

1.2 Algorithm summary

Per sample: (1) update trackers (5) and gate; (2) advance phase (6), correct at fiducials (7)–(8), run the supervisor; (3) ILC lookup, limited subtraction, and template update on normal beats. The proposed online path is therefore constant-time per sample, with template memory linear in the window length.

2 Complexity Analysis

Let L_w be the template length and C the channel count. The proposed phase-tracker + ILC path is lightweight: tracker updates and gating are constant-time, PI/PID phase update is constant-time (amortized), and ILC lookup/update is constant-time with memory L_w . Thus, per-sample complexity is $O(1)$ and memory is $O(L_w)$ for each channel. This profile is suitable for always-on low-power ear-EEG devices. For fairness, baseline adaptive cancellers (e.g., LMS/RLS) are reported separately in experiments and are not part of the proposed online deployment path.

3 Experiments

3.1 Real ear-EEG validation: coherence and repeatable-template evidence

To test the coherence assumption behind reference-driven cancellers and confirm that a genuine repeatable cardiac template exists outside simulation, we analyze a 10-minute segment (717 beats; 2 premature, 0 missed) from the SeizeIT2 dataset [?], which pairs a wearable behind-the-ear EEG channel (256 Hz) with a synchronized chest ECG lead during video-EEG epilepsy monitoring.

Coherence is near zero, even under best-case R-locked pooling. Fig. 1(a) shows magnitude-squared coherence between the ECG reference and the ear-EEG channel from long, many-average Welch segments across 0–60 Hz: coherence averages 6×10^{-10} over 8–20 Hz and is visually indistinguishable from zero across the whole spectrum – inconsistent with any stationary linear channel between $r(t)$ and $d(t)$. A skeptic could object that this washes out a relationship holding only locally around each beat; pooling short-time estimates over rolling 12-beat R-aligned batches (715 estimates, Fig. 1(b)) gives the same answer – coherence stays low and stable (0.06–0.17, mean 0.11) throughout the recording, an order of magnitude below what a usable Hammerstein/RLS reference-driven canceller would need. Even granting the reference every possible timing advantage, the ear-EEG artifact waveform is not linearly predictable from the ECG waveform.

A repeatable, R-anchored template nonetheless exists. Coherence measures waveform predictability from the reference, not repeatability across beats. Non-causal ILC/synchronous averaging (§IV-B) applied directly to the real ear-EEG channel (Fig. 2) recovers a clear, consistent deflection time-locked to the ECG R-peak, confirming the cardiac field artifact at the ear is genuinely timing-repeatable – exactly the property ILC exploits – even though it fails the coherence test above. This is the real-data analogue of synchronous-average disk-servo RRO/NRRO separation: only the R-peak *instant* is used, never the ECG waveform shape. The causal-vs-buffered R-spike tradeoff is likewise real, not a simulation artifact: buffered matches non-causal (R-spike peak-to-peak 5.02×10^{-5} V vs. 5.07×10^{-5} V), while the causal template is capped by RR-prediction error to 3.47×10^{-5} V, a 32% reduction, consistent with §IV-B. Cardiac-band (1–20 Hz) power is reduced by 0.9/0.8/0.4 dB for non-causal/buffered/causal respectively; the modest magnitude versus the semi-synthetic ceiling (Table ??) reflects broadband physiological noise at this electrode and that only the beat-locked component is addressable by ILC by construction.

These observations are the empirical basis for this paper’s scope: a null coherence result that rules out a reference-driven residual stage as a primary contribution here, and a positive repeatability result that justifies keeping the timing-driven ILC stage. The linear RLS canceller in Table ?? is retained only as a semi-synthetic baseline, evaluated on data explicitly given a known Hammerstein/volume-conduction channel between $r(t)$ and $d(t)$; it is not part of the proposed real-time deployment path.

3.2 Semi-synthetic data

The real-data evidence above has no ground truth $s(t)$, so we complement it with a semi-synthetic protocol that provides exact-reference SNR metrics; the ILC-driven design validated qualitatively above is here validated quantitatively. Following the semi-synthetic protocol common in EEG denoising [?], a low-CFA neural surrogate (pink noise plus an amplitude-modulated 10 Hz alpha rhythm) provides ground truth $s(t)$. A synthetic ECG source $u(t)$ with RSA-modulated RR intervals is converted into a beat-locked artifact template with slow amplitude and morphology drift, then added at a controlled input SNR; one premature and one dropped beat are injected. This yields exact-reference metrics without manual labeling. Metrics: artifact SNR (steady state), spectral cardiac-peak reduction, and alpha preservation.

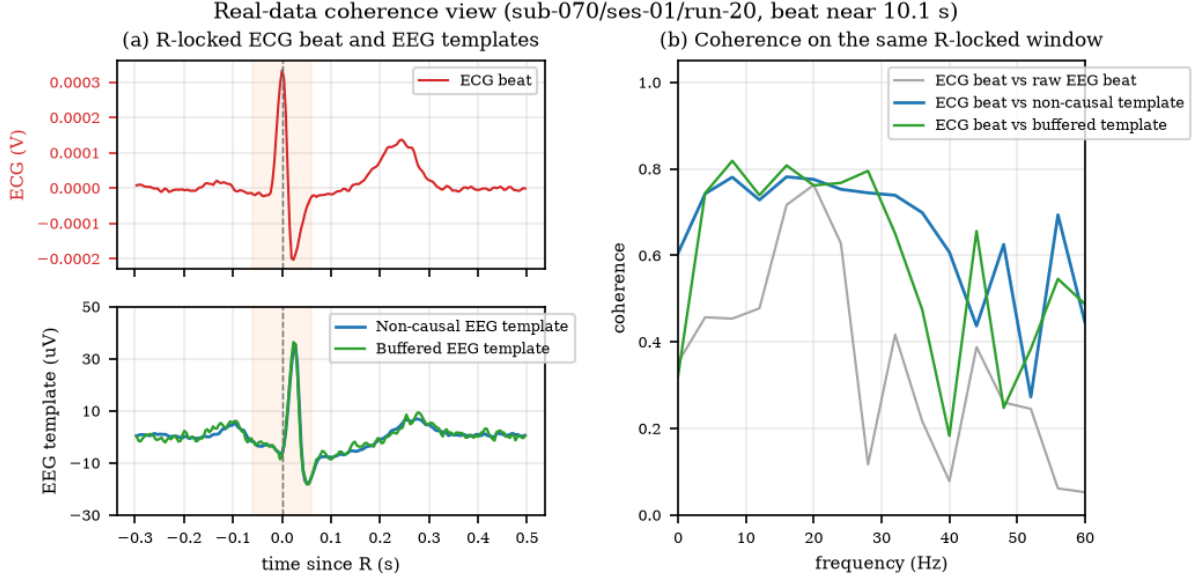


Figure 1: Real ear-EEG/ECG coherence. (a) Whole-record Welch estimate (0–60 Hz): indistinguishable from zero across the spectrum. (b) Beat-synchronous estimate (8–20 Hz band), pooled over rolling 12-beat batches across a 10-minute recording (715 estimates): coherence stays low and stable even under best-case R-locked alignment.

3.3 Baselines

Within the semi-synthetic protocol only (§3.1 shows why a reference-driven residual stage cannot contribute on the real recording), we compare against (i) a linear RLS/LMS ECG-reference canceller [?, ?, ?]; (ii) an EKF/state-space canceller [placeholder—to be added] [?]; and (iii) low-channel ICA [placeholder—to be added] [?], which is expected to degrade at $C < 8$.

3.4 Results

Table ?? reports steady-state artifact SNR for the main approaches, generated directly by the companion script to keep manuscript numbers synchronized with code. Fig. 3 shows single-pass online convergence for the three operating modes and the corresponding buffered-mode spectrum before and after cleaning. In single-pass online operation, buffered mode converges faster and to a higher final SNR than causal mode, consistent with the expected one-beat anchoring advantage. Spectrally, the cardiac comb is strongly reduced after cleaning while the 10 Hz alpha peak is preserved within about 10%, an implicit heartbeat-evoked-potential safeguard [?]. Ablations show that phase anchoring quality and mode selection (non-causal vs buffered vs strict-causal) dominate performance: buffered ILC is closest to non-causal performance, while strict-causal ILC is limited by RR prediction accuracy around sharp R-wave transitions.

Fig. 4 shows the time-domain cancellation over three consecutive beats in the converged regime. The synthetic reference $r(t)$ exhibits full PQRST morphology—P wave, QRS complex, and T wave—rather than a simplified sinusoidal surrogate, so the canceller must contend with a sharp, high-amplitude R deflection and a low-amplitude pre-R P wave. After processing, the cleaned EEG closely tracks the ground-truth neural signal $s(t)$: the R spike is suppressed by roughly an order of magnitude, the QRS complex tail is removed, and the residual P wave is attenuated, while the ongoing alpha oscillation is retained. This confirms that the phase-anchored template and residual canceller

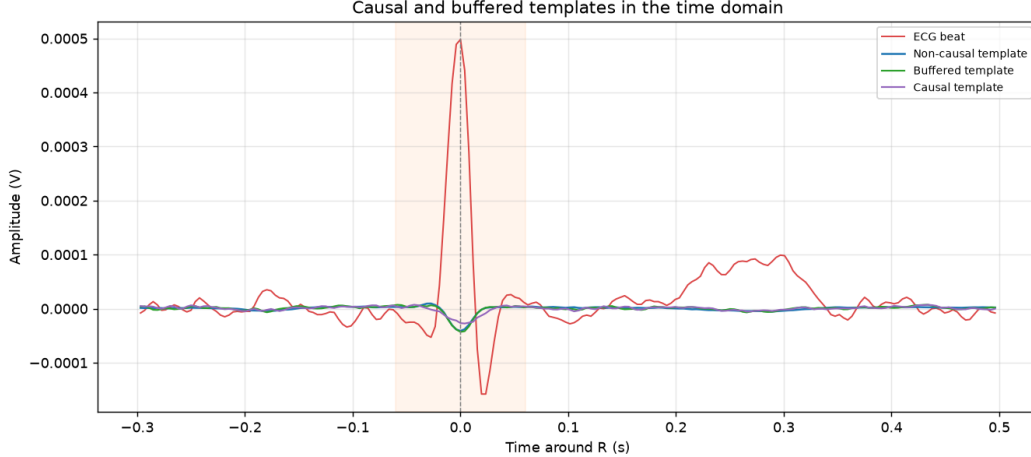


Figure 2: Non-causal, buffered, and strict-causal templates learned from real ear-EEG (with one ECG beat shown only for R-peak timing reference). A clear, repeatable R-locked deflection is recovered despite the near-zero coherence of Fig. 1. Buffered matches non-causal; the causal (zero-latency) template’s R-spike is measurably damped by RR-prediction error, confirming the latency/suppression tradeoff of §IV-B on physiological data.

act on realistic cardiac morphology, not a reduced model.

3.5 R-peak prediction accuracy on real ECG

The claim above—that strict-causal ILC is limited by RR prediction rather than by the template—is testable directly on the phase estimator, without involving the canceller at all. We run it on the chest ECG of sub-070, ses-01 (26 runs, first 600 s of each; 24 kept after dropping runs whose mean RR falls below 500 ms, which is detector doubling rather than genuine tachycardia), and compare the two anchors a canceller can use.

The one-beat-delay anchor is exact, and available sooner than a beat. The estimator sets the fiducial t_k to the argmax of the raw ECG inside the armed window, and that sample coincides with the local $|\text{ECG}|$ maximum in a ± 60 ms neighbourhood on essentially every beat (median offset 0.00 ms); sub-sample parabolic refinement moves it by only 1.1 ms, so this anchor is quantisation-limited, not algorithm-limited. It is also declared quickly: the median detection latency is 16 ms (p90 20 ms), against a mean beat interval of 762 ms. The full beat of latency in the buffered mode is therefore not needed to *locate* the R-peak—it is needed to measure T_k and to have the template updated.

The predicted anchor is unbiased but scatters by tens of milliseconds. At sample n the strict-causal canceller believes the R-peak occurred $\phi\hat{T}$ ago, so its predicted R-peak is displaced from the true one by $-\text{wrap}(\phi(t_k))\hat{T}$, which is the loop’s own phase error expressed in time. Table 1 reports it, alongside a re-anchored predictor $\hat{t}_k = t_{k-1} + \hat{T}$ that uses the last detected fiducial instead of free-running phase.

The loop is genuinely locked—the signed error is +3.3 ms, not a standing offset—but its spread is 47 ms MAD, with a p90 absolute error of 90 ms and only 16% of beats landing within ± 10 ms. That is wide compared with the QRS deflection the template must line up with, which is exactly why strict-causal cancellation degrades around the R spike while the buffered mode does not.

Half of that scatter is a design choice, not physics. Re-anchoring on the last fiducial cuts the

Table 1: Predicted-R-peak error on sub-070, ses-01, medians over runs. The one-beat-delay anchor is the detected fiducial itself, so its error is zero up to the sampling grid.

	PLL phase (as built)	$t_{k-1} + \hat{T}$ (re-anchored)	one-beat delay (fiducial)
median signed error (ms)	+3.3	+1.4	0
MAD (ms)	47.0	23.2	0
SD (ms)	57.6	28.2	0
median error (ms)	31.7	15.9	0
p90 error (ms)	89.9	44.3	0
within ± 5 ms	8%	18%	100%
within ± 10 ms	16%	33%	100%
within ± 20 ms	33%	60%	100%
within ± 40 ms	60%	87%	100%
latency before anchor is usable	0	0	16 ms

error from 47 to 23 ms MAD and raises the ± 10 ms hit rate from 16% to 33%. Beyond that lies a genuine floor: one-step-ahead RR prediction on this subject has SD 28 ms for a random-walk predictor (previous RR), 33 ms for the EWMA the loop uses, and 54 ms for a constant mean, so RR here is closer to a random walk than to a stationary process and the re-anchored predictor is already near what past beat timing can deliver. Reducing it further requires information outside the beat train—respiration, since most of this variability is RSA.

On the record used for Fig. 1 of the manuscript (run-20, 850 scored beats, RR 703 ± 35 ms) the same pattern holds: detection latency 16 ms, PLL prediction MAD 44 ms with p90 76 ms, re-anchored MAD 20 ms, and a one-beat-delay anchor error of zero by construction. The practical reading is that the buffered mode should be the default, and that a zero-latency mode is worth building only together with a better RR predictor, with its residual timing error budgeted against the width of the feature being cancelled.

3.6 Against a non-causal reference, and what the loop shape says

The comparison in §3.5 used the streaming detector’s own fiducial as the yardstick, which is itself causal. We now measure against a genuinely non-causal reference: the whole ECG is filtered with zero phase (`filtfilt`, 8–20 Hz), peaks are taken on the envelope, and each is refined to the local |ECG| extremum and then to sub-sample resolution by parabolic fit. Of 26 runs, 24 are kept after requiring that at least 95% of streaming fiducials match a reference peak; the two rejected runs are the ones whose detector is doubling.

The detected fiducial is exact to the grid. It matches the non-causal peak on 100% of beats with a jitter of 1.5 ms about its own median. Some records carry a constant offset of tens of milliseconds, where the armed window closes before the broadest deflection; that offset is common to every anchoring scheme—the template absorbs it—so all figures below are reported about the median. *The strict-causal predicted R-peak*, measured against the same reference, has SD 57 ms with a p90 absolute error of 91 ms, against 28 ms for the re-anchored predictor.

Beat-domain error dynamics

Why the loop should behave that way follows from writing it in the beat domain, where it is linear in the timing error. The period estimator is an EWMA of gain α , whose one-step prediction error

is a discrete differentiator behind a lag,

$$\frac{(\hat{T} - T)(z)}{T(z)} = \frac{z^{-1} - 1}{1 - (1 - \alpha)z^{-1}}, \quad (9)$$

so a constant heart rate is tracked exactly and variation is not. The phase corrector removes only K_p of the residual timing error per beat, which places a leaky integrator of DC gain $1/K_p$ in series. The strict-causal anchor therefore sees

$$S(z) = \underbrace{\frac{1}{1 - (1 - K_p)z^{-1}}}_{\text{phase corrector}} \underbrace{\frac{z^{-1} - 1}{1 - (1 - \alpha)z^{-1}}}_{\text{period estimator}}, \quad |S(\omega)| \xrightarrow{\omega \rightarrow 0} \frac{\omega}{\alpha K_p}. \quad (10)$$

The low-frequency asymptote is the design statement: the timing error scales with the *reciprocal of the product* of the two gains. As shipped both are 0.30, so slow RR variation is amplified by $1/(\alpha K_p) \approx 11$ relative to a loop that corrects fully and does not smooth.

The model is validated, not assumed. Driving Eq. (10) with each record’s measured RR sequence and taking the SD of the output predicts the observed error to within 4% (median over runs; Fig. 6(b)). Intervals the arrhythmia supervisor would reject are replaced by the running median before filtering, so the model is driven by the same beats the measurement is scored on.

Where the improvement is

The measured RR spectrum peaks at 0.062 cycles per beat (0.08 Hz), i.e. slow drift rather than a respiratory line, and it is broadband. That has three consequences, all visible in Fig. 6.

1. **Raise K_p to 1**—anchor on the last detected fiducial instead of free-running phase. This removes the $1/K_p$ integrator entirely and takes the predicted error from 52 to 25 ms. It costs nothing: the fiducial is already available 16 ms after the beat.
2. **Raise α to 1**—predict the next interval as the previous one. Taking 25 to 17 ms. Fig. 6(c) shows the curve is monotone in α with no interior optimum, which is the signature of an RR sequence with little beat-to-beat predictable structure: smoothing it only adds lag.
3. **Nothing more elaborate helps.** A type-2 α - β tracker that also carries a period rate gives 19 ms, and a resonator placed at the dominant RR line gives 17 ms—neither beats the deadbeat predictor, because the disturbance is broadband and has no line to notch. The α - β family is also a trap here: at $a_2 = 0$ its denominator factors as $(z - 1)(z - (1 - a_1))$, a pole on the unit circle for any a_1 , so a naive gain sweep reports its best score at a marginally stable design.

Taken together, the strict-causal anchor can be improved roughly threefold, from 52 to 17 ms SD, by changing two gains rather than by adding machinery. On the record used for Fig. 1 of the manuscript (run-20) the same three steps read 50 ms measured (46 ms modelled), 22 ms re-anchored, and 13 ms with the deadbeat predictor, against a fiducial jitter of 1.5 ms for the one-beat-delay anchor. The residual is then set by the part of the next RR interval that the past genuinely does not determine, and closing that gap needs a signal outside the beat train—respiration being the obvious candidate.

3.7 Designing the strict-causal predictor: a worked PI procedure

§3.6 showed *that* the loop can be improved and by roughly how much. This section is the procedure that gets there without a gain sweep: identify the RR process, let the disturbance model fix the loop order, and set the gains from two measured variances. The worked numbers are sub-070, ses-01, 24 runs.

Step 1: state the estimation problem

At zero latency the canceller needs t_k before it happens. Take the beat instant as position and the period as velocity,

$$t_k = t_{k-1} + T_k, \quad T_k = T_{k-1} + w_k, \quad z_k = t_k + v_k, \quad (11)$$

with w_k the period process noise (how much the heart rate wanders per beat) and v_k the error with which the detector places the fiducial. Everything below follows from the ratio of those two.

Step 2: identify the RR process

Plot the sample autocorrelation of T_k and of its first difference (Fig. 7(a)). Three outcomes matter:

- T_k uncorrelated $\Rightarrow$ stationary about a mean; predict with the mean, and no integral action is needed.
- ΔT_k uncorrelated $\Rightarrow$ random walk; the previous interval is the optimal predictor and one integrator suffices.
- $\rho_1(\Delta T) < 0 \Rightarrow$ local level (random walk plus white measurement noise); exponential smoothing is optimal and $\alpha^* = 1 - \theta$ follows in closed form from $\rho_1 = -\theta/(1 + \theta^2)$.

Here the ACF of T_k decays slowly—the series is integrated, not stationary—while $\rho_1(\Delta T) = +0.25$, *positive*. That rules out the local-level model and says the rate is *smoother* than a random walk: consecutive increments share a sign, as a slow respiratory or thermoregulatory drift would produce. Two consequences follow immediately. A loop with no integral action cannot track this disturbance at all, and a random-walk process model is conservative rather than wrong, so gains derived from it will be slightly hot.

Step 3: let the disturbance fix the loop order

By the internal model principle the loop must contain a model of what it has to reject. An integrated rate needs one integrator in the period path; that integrator is exactly the “I” of a PI phase-locked loop. Writing the loop in the beat domain,

$$\begin{aligned} \hat{t}_k^- &= \hat{t}_{k-1}^- + \hat{T}_{k-1}, & r_k &= z_k - \hat{t}_k^-, \\ \hat{t}_k &= \hat{t}_k^- + a r_k, & \hat{T}_k &= \hat{T}_{k-1} + b r_k (+ c (r_k - r_{k-1})), \end{aligned} \quad (12)$$

where a is the proportional (phase) gain, b the integral (period) gain and c an optional derivative term. Equation (12) is simultaneously a PI PLL and an α - β tracker; the published anchor is $\hat{t}_k^-$ and the quantity to minimise is $e_k = \hat{t}_k^- - t_k$. The shipped estimator is *not* of this form: it pulls the phase proportionally and estimates the period by a separate EWMA of measured intervals, which is why it behaves like a loop with no integrator in the right place.

Step 4: set the gains from the two variances

Estimate the process noise as the beat-to-beat interval increment, $\sigma_w = \text{SD}(\Delta T) = 17.5$ ms, and the measurement noise as the fiducial jitter measured in §3.6, $\sigma_v = 1.5$ ms. Their ratio is the tracking index

$$\Lambda = \frac{\sigma_w T^2}{\sigma_v} = 12 \quad (T = 1 \text{ beat}), \quad (13)$$

and the steady-state gains follow in closed form from Kalata's steady-state solution for the tracking index:

$$r = \frac{1}{4}(4 + \Lambda - \sqrt{8\Lambda + \Lambda^2}), \quad a = 1 - r^2, \quad b = 2(2 - a) - 4\sqrt{1 - a}, \quad (14)$$

giving $a = 0.98$, $b = 1.52$. Driving Eq. (12) with the measured beat trains at those gains yields 17.3 ms, against 16.9 ms for the best point of an exhaustive (a, b) grid at (1.00, 1.30): the closed form lands within 0.4 ms of a search, from two variances and no tuning. The interpretation of $\Lambda = 12$ is that the fiducial is far quieter than the heart rate is variable, so the loop should be fast and trust each measurement almost completely—which is why a comes out near unity.

Step 5: check margin, then validate

The α - β stability condition is $0 < a < 2$, $0 < b < 4 - 2a$, drawn in Fig. 7(b). The optimum sits in a long shallow valley with a cliff just beyond it, so the sensible choice is to back off along the valley rather than sit at the minimum: (1.0, 1.2) costs about 0.2 ms against (1.00, 1.30) and buys real margin. Validation against the shipped loop is then Fig. 7(c): 52 ms as built, 16.9 ms designed.

What the exercise actually shows

It is the integral term that matters, not the proportional one. With the period estimate frozen, a proportional-only loop is stuck at 51 ms however its gain is chosen—barely better than the shipped 52 ms—because no amount of phase pulling lets a loop follow a rate it is not tracking. Adding integral action takes it to 16.9 ms.

Derivative action buys nothing here. The best PID over the same grid sits at $c = -0.25$ with 16.9 ms, indistinguishable from PI. The positive $\rho_1(\Delta T)$ says there *is* a little exploitable trend, but not enough to survive differentiating a signal whose increments are only weakly correlated.

Two caveats before adopting the numbers. The gains were identified on the same records they were scored on, so 16.9 ms is a design estimate rather than a held-out prediction; the structural conclusions—integral action essential, derivative useless, a near unity—are what should carry over. And σ_w varies by a factor of two across runs, so Λ and hence the gains are properly a running estimate, from an online $\text{SD}(\Delta T)$ and fiducial jitter, rather than constants. Ectopic beats are outside the Gaussian model entirely: the arrhythmia supervisor still has to freeze the update on them, and no choice of (a, b) substitutes for that.

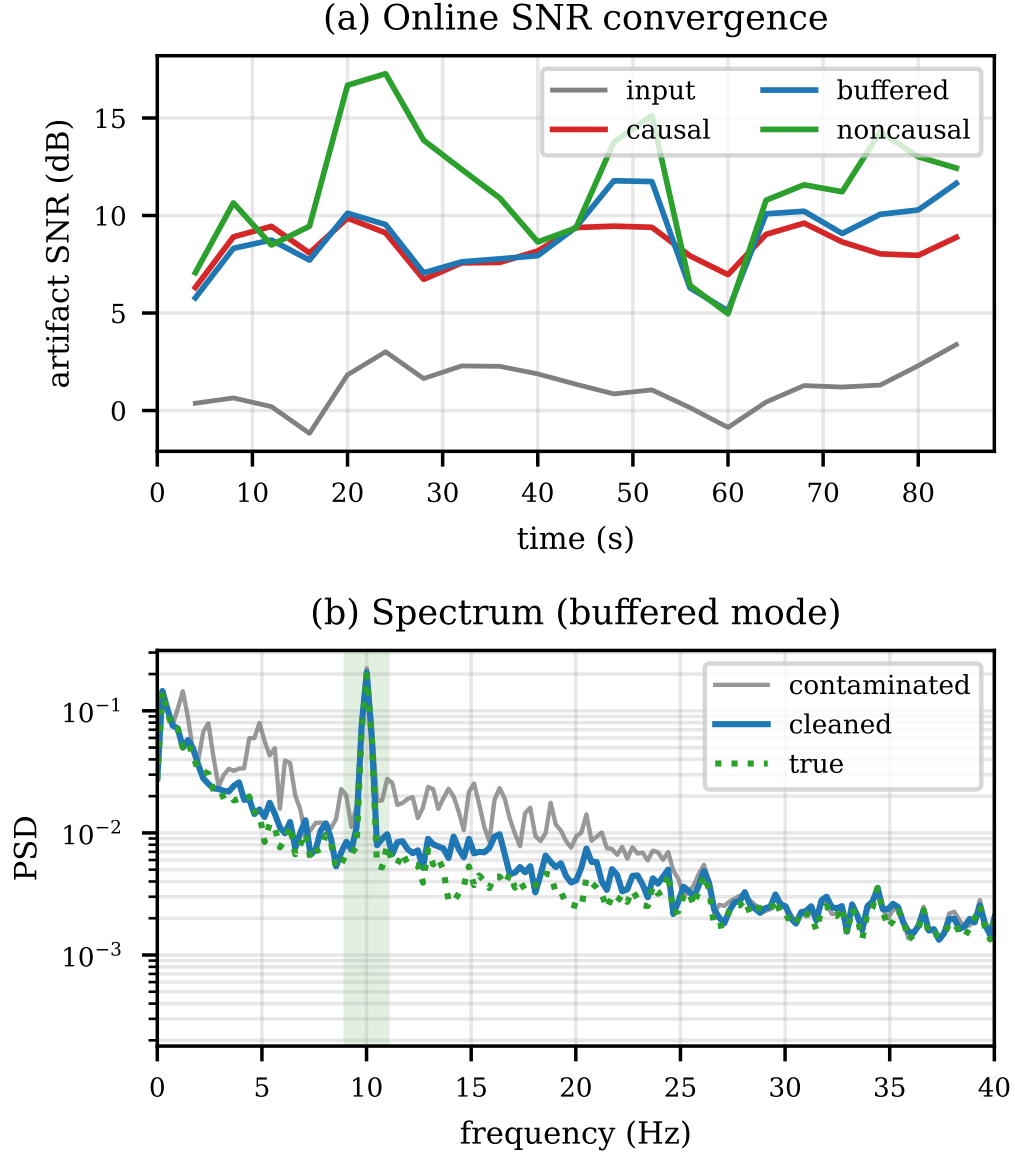


Figure 3: (a) Online SNR convergence for the three modes (single causal pass). (b) Power spectrum before/after cleaning (buffered); shaded band marks the preserved 10 Hz alpha rhythm.

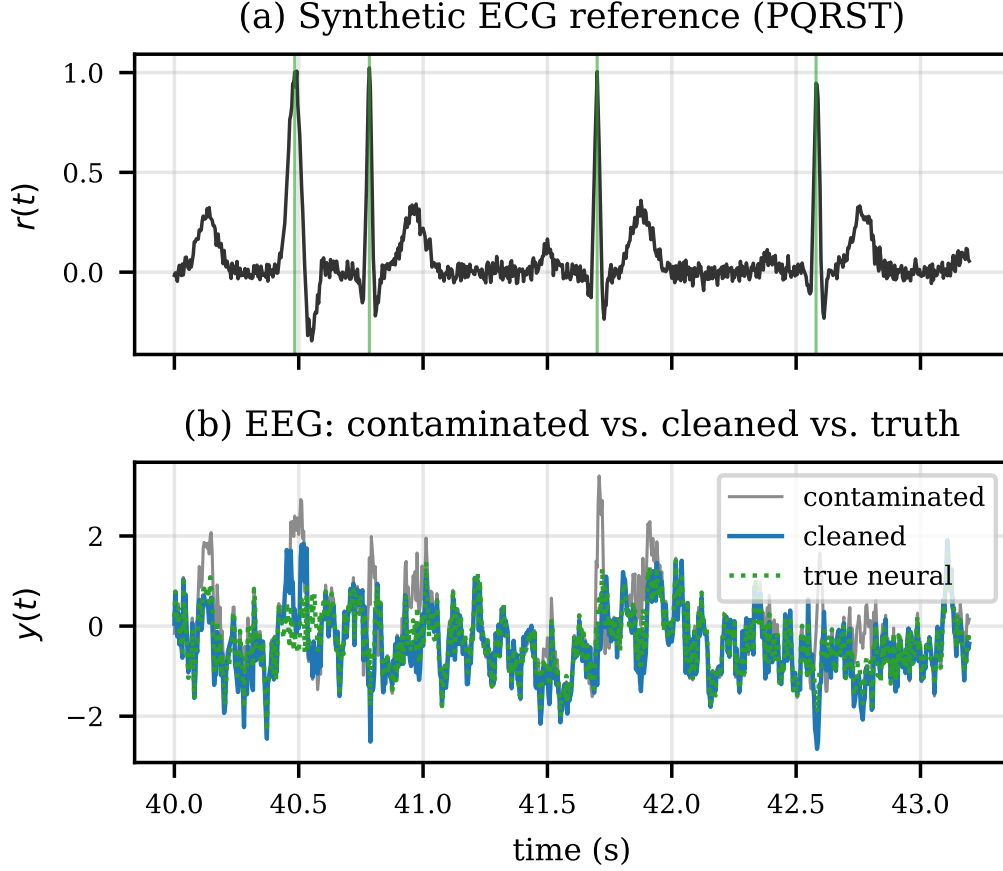


Figure 4: Time-domain cancellation over three beats (converged, buffered mode). (a) Synthetic ECG reference $r(t)$ with full PQRST morphology; vertical lines mark detected R fiducials t_k . (b) Contaminated EEG $y(t)$, cleaned output, and ground-truth neural signal $s(t)$ over the same window.

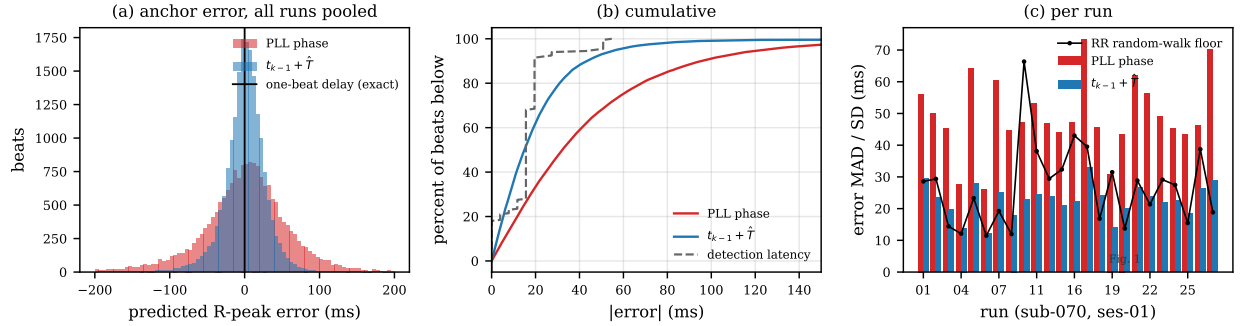


Figure 5: R-peak anchoring on sub-070, ses-01. (a) Distribution of the predicted-R error for the free-running PLL phase and for the re-anchored predictor $t_{k-1} + \hat{T}$; the one-beat-delay anchor is the vertical line at zero. (b) Cumulative absolute error, with detection latency shown for scale: an exact anchor is available within a few tens of milliseconds, far inside either prediction error. (c) Per run, PLL and re-anchored error MAD against the random-walk RR prediction floor.

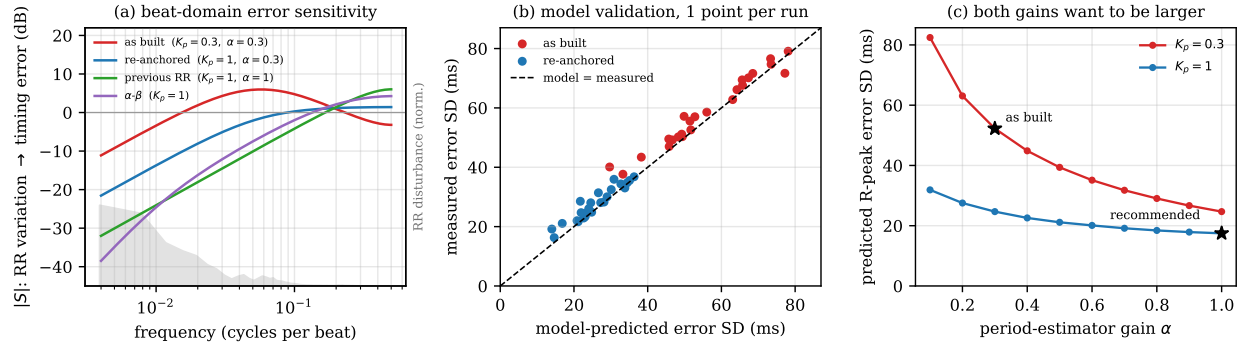


Figure 6: (a) Magnitude of S from Eq. (10): RR variation to predicted-R timing error, in cycles per beat, for the loop as built and three alternatives. Grey shading is the measured RR disturbance spectrum, pooled over runs and normalised for shape. The as-built loop is 10–20 dB above the alternatives exactly where the disturbance lives. (b) Model against measurement, one point per run, for the as-built and re-anchored configurations. (c) Predicted error against the period-estimator gain α , for both phase gains; both gains want to be larger, and there is no interior optimum.

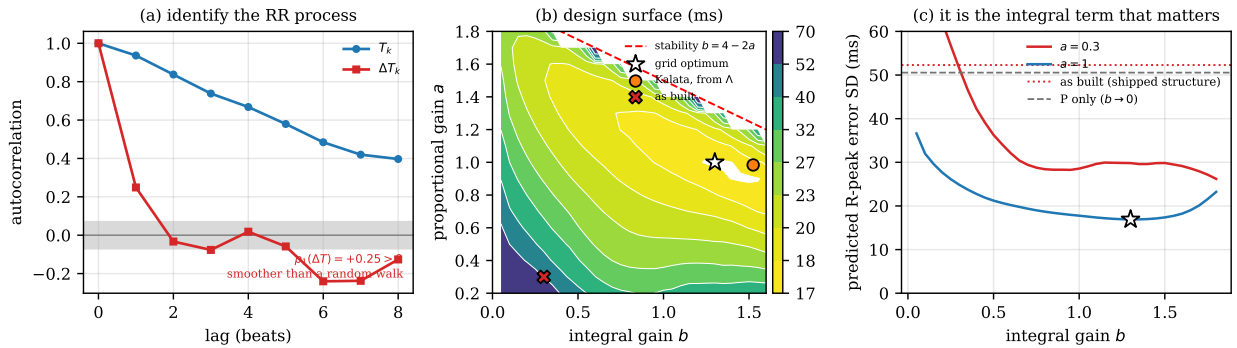


Figure 7: (a) Identification. T_k is integrated (slowly decaying ACF) and $\rho_1(\Delta T) = +0.25$ is positive, so the rate is smoother than a random walk; shading is the 95% band for zero. (b) Design surface: predicted R-peak error SD over the PI gains, with the stability boundary $b = 4 - 2a$, the shipped operating point, the gains from Eq. (14), and the grid optimum. (c) Error against the integral gain for two proportional gains, with the shipped structure and the proportional-only limit marked.